TENDER SPECIFICATION DOCUMENT: ADVANCED SPECTROSCOPY SYS

SECTION A: GENERAL TECHNICAL SPECIFICATIONS (PARALLEL COLUMNS)

1. Mechanical & Structural Subsystem

• Overall system footprint shall not exceed 1200 x 800 x 1600 mm (W x D x H);

• Total instrument weight: maximum 450 kg including vibration dampeners;

• Enclosure protection rating: IP54 compliant for industrial dust resistance;

• Structural frame: anodized aluminum profile with active leveling castors;

• Acoustic noise level: < 55 dBA at 1 meter during normal operation;

2. Electrical & Power Subsystem

• Input operating voltage: 230 V AC +/- 10%, 50 Hz single phase utility power;

• Total power consumption: maximum 2.5 kW under full cryogenic load;

• Power conversion efficiency: >= 92% at rated nominal load;

• Emergency power off (EPO): dual circuit break interlock switch provided;

• Integrated online UPS: 3 kVA true online double-conversion backup system;

SECTION B: SUBSYSTEM DETAILED SPECIFICATIONS (3-COLUMN MATRIX)

Optical Subsystem

• Excitation laser: 532 nm +/- 0.5 nm wavelength;

• Output optical power: 100 mW continuous wave;

• Spectral resolution: better than 0.5 cm-1;

• Beam divergence: < 1.0 mrad full angle;

Environmental Chamber

• Ambient temperature range: 18 °C to 28 °C;

• Relative humidity: 20% to 75% non-condensing;

• Thermal stabilization time: < 30 minutes from cold start;

• Heat dissipation: < 1.2 kW into ambient laboratory air;

Data & Control System

• Communication bus: Gigabit Ethernet (1000Base-T);

• Data acquisition ADC: 16-bit at 100 MSPS sampling rate;

• Controller OS: Linux Ubuntu 22.04 LTS 64-bit;

• Export format support: CSV, JSON, HDF5 and ASCII formats;